

EXPT NO:5B	DATA VISUALIZATION

AIM:

To visualize the **AI Impact on Jobs 2030 dataset** using Python and create different charts to understand the distribution of job-related and AI-related attributes and identify important patterns in the dataset

PROCEDURE:

1. Start the program.
2. Import Pandas, Matplotlib, and Seaborn libraries.
3. Load the AI Impact on Jobs CSV dataset.
4. Display the first few records using `head()`.
5. Select the numerical columns from the dataset.
6. Create a bar chart to compare jobs based on their AI Replacement Risk.
7. Create a line chart to analyze AI Replacement Risk across different jobs.
8. Create a pie chart to show the distribution of jobs based on AI Tool Usage.
9. Create a box plot to visualize the distribution of Average Salary.
10. Create a histogram to show the distribution of AI Replacement Risk.
11. Add suitable titles, axis labels, and legends.
12. Display all the visualizations.
13. Analyze the graphs and identify important patterns related to AI impact on jobs.
14. Stop the program

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("AI_Impact_on_Jobs_2030.csv")

print(df.head())

num_cols = df.select_dtypes(include="number").columns
print("Numerical columns:", num_cols.tolist())
```

```
plt.figure(figsize=(8,5))
df["AI_Replacement_Risk"].head(10).plot(kind="bar")
plt.title("AI Impact on Jobs - Bar Chart")
plt.xlabel("Job")
plt.ylabel("AI Replacement Risk")
plt.show()
```

```
plt.figure(figsize=(8,5))
plt.plot(df["AI_Replacement_Risk"].head(20), marker="o")
plt.title("AI Impact on Jobs - Line Chart")
plt.xlabel("Job")
plt.ylabel("AI Replacement Risk")
plt.grid()
plt.show()
```

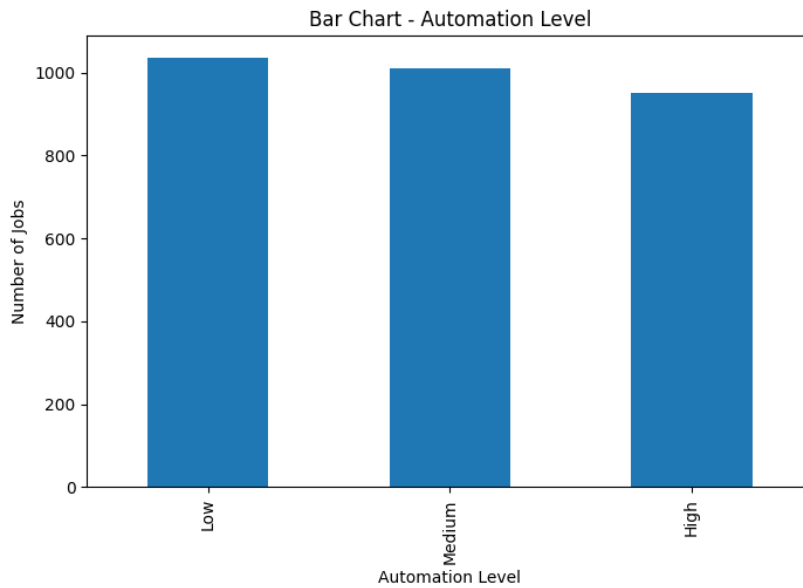
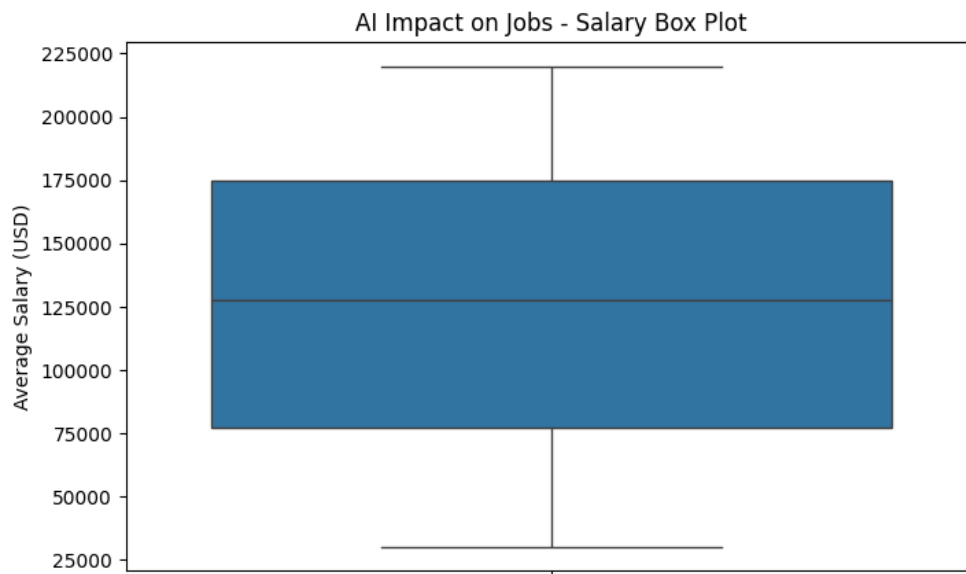
```
plt.figure(figsize=(7,7))
df["AI_Tool_Usage"].value_counts().plot(
    kind="pie",
    autopct="%1.1f%%"
)
plt.title("AI Impact on Jobs - AI Tool Usage")
plt.ylabel("")
plt.show()
```

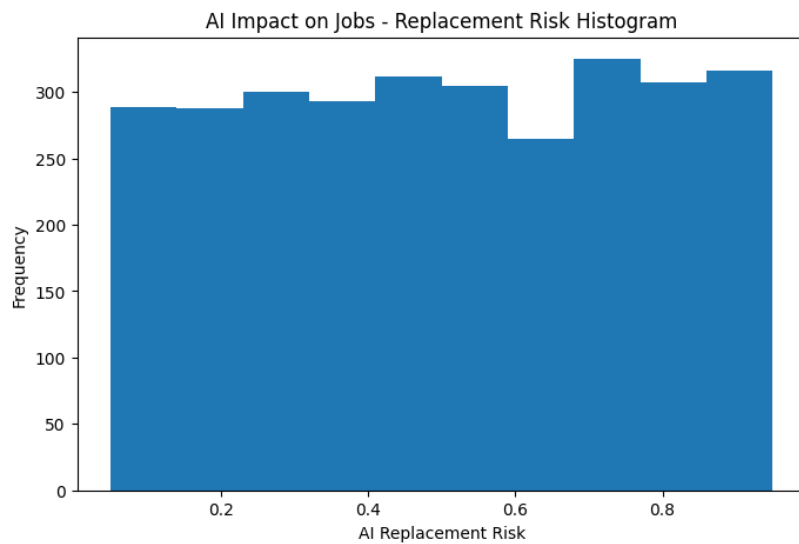
```
plt.figure(figsize=(8,5))
sns.boxplot(y=df["Average_Salary_USD"])
plt.title("AI Impact on Jobs - Salary Box Plot")
plt.ylabel("Average Salary (USD)")
plt.show()
```

```
plt.figure(figsize=(8,5))
plt.hist(df["AI_Replacement_Risk"], bins=10)
```

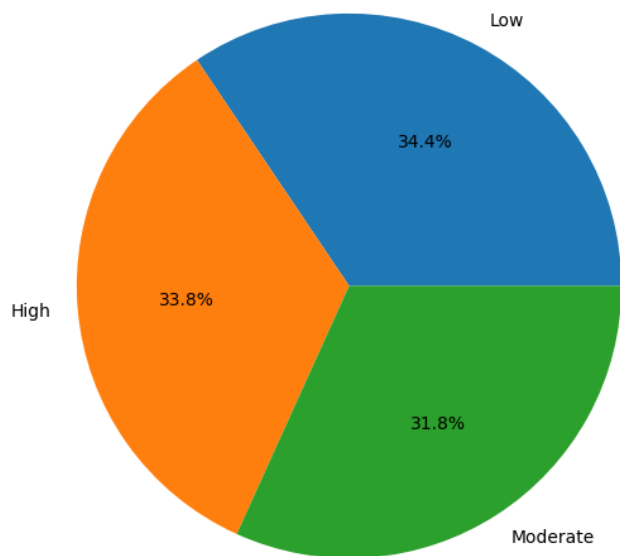
```
plt.title("AI Impact on Jobs - Replacement Risk Histogram")
plt.xlabel("AI Replacement Risk")
plt.ylabel("Frequency")
plt.show()
```

OUTPUT:





AI Impact on Jobs - AI Tool Usage



RESULT:

Thus, the Data Visualization of the AI Impact on Jobs 2030 dataset using Python libraries such as Pandas, Matplotlib, and Seaborn was implemented successfully, and various charts were generated to identify important patterns related to AI replacement risk, AI tool usage, salary, and job impact.